運輸署
Transport Department

Survey on Goods Vehicle Trip Characteristics 2011 (GVTCS 2011)

Final Report



December 2014

ARUP

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Acronyms

CBTS	Cross Boundary Travel Survey
CPI	Consumer Price Index
CTS	Comprehensive Transport Study
GDP	Gross Domestic Product
GV	Goods Vehicle
GVCS	Goods Vehicle Composition Survey
GVIS	Goods Vehicle Owner and Driver Interview Survey
GVTCS 2003	Survey on Goods Vehicle Trip Characteristics 2003
GVTCS 2011	Survey on Goods Vehicle Trip Characteristics 2011
GVTGS	Goods Vehicle Trip Generation Survey
GVW	Gross Vehicle Weight
HGV	Heavy Goods Vehicle
HKSAR	Hong Kong Special Administrative Region
LGV	Light Goods Vehicle
MGV	Medium Goods Vehicle
NT	New Territories
PlanD	Planning Department
SG	Special Generator
SGIS	Special Generator Interview Survey
TD	Transport Department
VALID	Vehicles and Drivers Licensing Integrated Data System
VJTS	Vehicle Journey Time Survey
VOC	Vehicle Operating Cost
VoT	Value of Time

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EXECUTIVE SUMMARY

E.1 Background

E.1.1 The Comprehensive Transport Study Model Enhancement – Feasibility Study (the Consultancy Study) was commissioned by Transport Department in July 2010, an integral part of which was the Survey on Goods Vehicle Trip Characteristics 2011 (GVTCS 2011) to collect up-to-date goods vehicle trip characteristics data and archive such data into a database. The database was subsequently used for subsequent re-calibration and enhancement of the goods vehicle trip model within the Comprehensive Transport Study (CTS) Model, for future transport planning. The last GVTCS was conducted in 2003 (GVTCS 2003).

E.1.2 The Consultancy Study defined the scope, data requirements and outline design of the GVTCS 2011. The planning, organisation and execution of the GVTCS 2011 surveys proposed by the Consultancy Study were undertaken under a separate service contract "Provision of Services for Conducting the Survey on Goods Vehicle Trip Characteristics 2011" (the Service Contract).

E.1.3 The GVTCS 2011 comprised five surveys:

- Goods Vehicle Owner and Driver Interview Survey (GVIS) – to obtain comprehensive information on the trip patterns of goods vehicles operating within the HKSAR. This is essential for re-calibrating and enhancing the goods vehicle trip model of the CTS Model and providing information for transport planning.
- Special Generator Interview Survey (SGIS) – to supplement the GVIS by collecting trip generation and characteristics data on goods vehicle movements to and from the "special generator"[1] (SG) sites.
- Goods Vehicle Trip Generation Survey (GVTGS) – to establish trip generation of different types of goods vehicles generated by different land use developments for various time periods on a normal weekday.
- Goods Vehicle Composition Survey (GVCS) – to collect traffic flow data for estimating the composition of goods vehicle traffic by vehicle type on major roads in Hong Kong at different times of a normal weekday. The data was used as a basis for validating the goods vehicle matrices developed from the GVIS and SGIS data.
- Vehicle Journey Time Survey (VJTS) – to obtain supplementary information on the average journey time taken by vehicles to traverse selected road links for reviewing the average operating speed of traffic in the road network.

E.1.4 For all the surveys, the classification of goods vehicles into five vehicle types as shown in **Table E.1** was adopted. Such classification is also consistent with those of the transport model setup.

Table E.1 Goods Vehicle Classifications

Vehicle Type	Description	Permitted Gross Vehicle Weight (tonnes)
Van-type LGV#	• LGV constructed with a fully enclosed body which is an integral part of the vehicle	≤ 5.5
Light Goods Vehicle (LGV)*	• Usually shorter than 6m • 2-axled light goods vehicle other than van-type LGV	≤ 5.5
Medium Goods Vehicle (MGV)*	• Usually longer than 6m • 2-axled with double rear tyres or 3-axled	> 5.5 to 24
Heavy Goods Vehicle (HGV)*	• Usually longer than 7m • Usually 4-axled	> 24 to 38
Container Vehicle#	• All articulated vehicles	≤ 44

* *Cap 374 - Road Traffic Ordinance*
\# *Cap 374A - Road Traffic (Construction and Maintenance of Vehicles) Regulations*

E.1.5 The fieldwork period and enumeration results of the respective surveys are summarised in **Table E.2**. All fieldwork was suspended during the Christmas and New Year holiday period.

[1] This refers to sites that generate significant goods vehicle movements with specific trip characteristics. For the purpose of GVTCS 2011, SG sites were defined to cover: Kwai Tsing Container Terminal, River Trade Terminal/ Mid-stream Site/ Private Wharf, Public Cargo Working Area, Wholesale Market, Air Cargo Terminal, Freight Forwarding Centre/ Logistic Centre/ Warehouse, Open Storage and Port Back-up Facility, Landfill Site, Barging Point and Hospital.

Table E.2 Fieldwork Period and Enumeration

Survey	Fieldwork Period	Enumeration
Goods Vehicle Owner and Driver Interview Survey (GVIS) *(Response Rate: 70%)*	10 Oct 2011 – 22 Feb 2012	4,470 Non-government vehicles + 121 government vehicles
Special Generator Interview Survey (SGIS)	27 Oct 2011 – 13 Jan 2012	80 sites involving 27,220 trip records (2-way)
Goods Vehicle Trip Generation Survey (GVTGS)	6 Oct 2011 – 20 Dec 2011*	219 sites
Goods Vehicle Composition Survey (GVCS)	6 Oct 2011 – 20 Dec 2011#	104 stations
Vehicle Journey Time Survey (VJTS)	6 Oct 2011 – 16 Jan 2012	29 routes

* Supplementary surveys were conducted at two sites on 17 April 2012.
Re-surveys were conducted at three sites on 7-8 February 2012.

E.1.6 The data collected in the surveys were then processed, expanded and adjusted based on independent control data.

E.1.7 The major findings of the surveys are summarised in the following paragraphs.

E.2 Goods Vehicle Owner and Vehicle Usage

Profile of Goods Vehicle Owners

E.2.1 The GVIS covered goods vehicles licensed with the Transport Department ("non-government vehicles") by random sampling as well as goods vehicles owned by the Hong Kong Government ("government vehicles").

E.2.2 Among the non-government goods vehicles, the largest proportion of their owners were engaged in the "Transportation, Storage, Postal and Courier Services" sector (55%). It was followed by "Construction" (15%) and "Wholesale, Retail and Import/Export Trade" (13%). "Manufacturing" industry accounted for only 4% of the goods vehicles, which dropped from 8% in 2003. About 2% of the goods vehicles were for non-commercial use, most of them being goods vans.

Goods Vehicle Usage

E.2.3 As at September 2011, the total goods vehicle fleet size in Hong Kong was 111,946 vehicles, comprising 110,660 non-government vehicles and 1,286 government vehicles. This was about 1% less than the total fleet size of 113,380 goods vehicles in GVTCS 2003.

E.2.4 The proportions of goods vehicles by vehicle type are illustrated in **Figure E.1**. Van-type LGVs accounted for the largest proportion of the goods vehicle fleet (39%) which increased from 34% in GVTCS 2003.

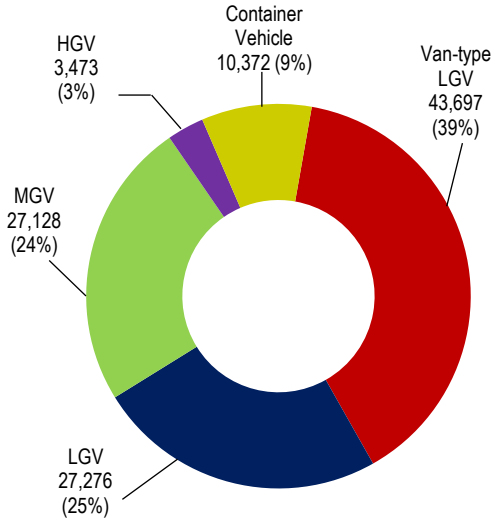


Figure E.1 : Goods Vehicle Fleet by Vehicle Type

E.2.5 The average numbers of working hours of drivers and delivery staff deployed for each goods vehicle per week are given in **Table E.3**. On average, 48 hours of drivers' and 39 hours of delivery staff's time were deployed in a week for each goods vehicle.

Table E.3 Average Numbers of Working Hours of Goods Vehicle Drivers and Delivery Staff Per Week

Vehicle Type	Average Number of Working Hours (1) Per Week	
	Driver	Delivery Staff
Van-type LGV	45.0	34.7
LGV	46.8	42.7
MGV	50.0	43.0
HGV	51.8	39.0
Container Vehicle	54.7	13.0
Overall	**48.0**	**39.3**

Note: (1) All goods vehicles excluding non-commercial use.

E.2.6 The average operating hours of the goods vehicles was estimated to be about 2,500 hours per year. This was marginally more than the 2,420 hours estimated in GVTCS 2003.

E.2.7 The overall average annual distance travelled by a goods vehicle including cross-boundary trips was 38,300 km, notably higher than the 31,800 km estimated in GVTCS 2003 which was confined to travel distances within HKSAR territory only. Refer to **Table E.4** below.

Table E.4 Average Annual Kilometrage Travelled by Goods Vehicle Type in 2003 and 2011

Vehicle Type	Annual Kilometres Travelled (km) (1)	
	2003	2011
Van-type LGV	30,300	27,700
LGV	29,900	31,200
MGV	32,500	46,400
HGV	39,800	46,900
Container Vehicle	36,900	78,900
Overall	**31,800**	**38,300**

Note: (1) The GVTCS 2011 covered the distances travelled outside the HKSAR for cross-boundary trips while the GVTCS 2003 were confined to that within the HKSAR territory only.

Overnight Parking of Goods Vehicles

E.2.8 The majority (95%) of goods vehicles were parked within Hong Kong. Only 5% of goods vehicles were parked in the Mainland, as compared to 10% in 2003.

E.2.9 When the goods vehicles were parked overnight in Hong Kong, most (71%) of them were parked in the New Territories, including rural Northwest NT (13%) and Kwai Chung (11%).

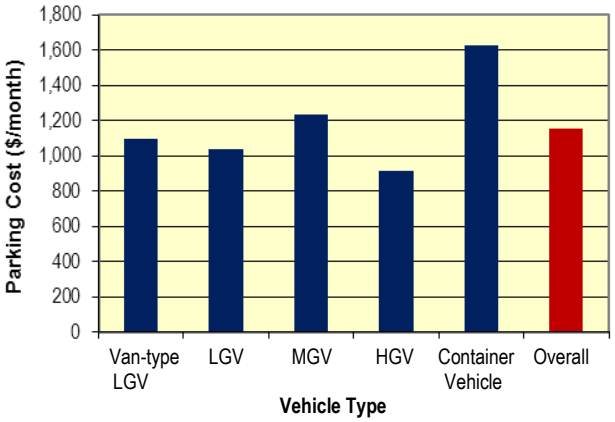
E.2.10 Referring to **Table E.5**, the vast majority (90%) of the goods vehicles were parked off-street, with monthly rental space (57%) being the most common type of parking, followed by designated rent-free space (18%) and self-owned space (8%).

Table E.5 Overnight Parking Arrangement in Hong Kong

Type of Parking Space		Proportion of Goods Vehicles
Off-street	Self-owned space	8%
	Rent-free space (designated)	18%
	Monthly rental space	57%
	Hourly/daily rental space	3%
	Designated loading/unloading space	*
	Non-designated	4%
On-street	Metered	4%
	Designated non-metered	*
	Non-designated	5%
Total		**100%**

* Less than 0.5%

E.2.11 The average monthly costs paid for parking different types of goods vehicles (excluding government vehicles) overnight in Hong Kong are illustrated in **Figure E.2**. Average overnight parking cost was $1,160/month.



(Base: Non-government goods vehicles)

Figure E.2 : Average Overnight Parking Costs Paid for Goods Vehicles

E.2.12 The lowest parking cost ($910/month) for HGVs could be attributed to the larger proportion of HGVs parked at non-designated parking spaces as compared to other vehicle types. All government goods vehicles were parked at designated off-street rent-free spaces, therefore they did not incur any parking cost.

Costs for Operating Goods Vehicles

E.2.13 Table E.6 summarises the annual expenses, itemised by type of expenses and vehicle type.

Table E.6 Annual Expenses Incurred in Operating Goods Vehicles

Expenses	GVan	LGV	MGV	HGV	CV
Parking Fee	20,370 (15,000)	19,770 (16,300)	23,270 (18,800)	15,660 (17,700)	28,300 (19,000)
Fuel Cost	40,510 (30,500)	60,390 (35,900)	112,270 (73,600)	171,070 (99,000)	194,070 (115,300)
Tyre Replacement/ Repair/ Maintenance Cost	10,810 (9,900)	16,680 (11,100)	30,130 (21,800)	51,710 (32,200)	44,550 (26,900)
Insurance Premium	5,390 (5,100)	6,150 (5,400)	9,070 (9,300)	13,080 (13,600)	13,990 (9,200)
Licence Fee	4,160 (4,300)	2,390 (2,400)	$4,670 (4,700)	4,690 (4,700)	4,690 (4,700)
Sub-total Operational Cost	**81,240 (64,800)**	**105,380 (71,100)**	**179,410 (128,200)**	**256,210 (167,200)**	**285,600 (175,100)**
Salary Costs	***157,000 (211,800)***	***178,500 (202,700)***	***208,260 (243,900)***	***199,270 (264,800)***	***185,940 (273,900)***
Annualised Vehicle Cost	***20,160* (18,700)**	***24,360* (25,200)**	***41,790* (42,400)**	***64,230* (63,000)**	***46,430* (62,700)**

Legend: "GVan" denotes "Van-type LGV"; CV denotes "Container Vehicle"

Note: 2003 figures are shown in ( ).

Monetary values above are expressed in nominal terms (i.e. money-of-the-day). Between 2003 and 2011, the Composite Consumer Price Index has increased by 18.2%.

E.2.14 Amongst the operating expenses of goods vehicles, salary costs constituted the highest proportion of the total, varying between 40% and 60% for different vehicle types. Fuel cost came second (17% - 41%). As compared to 2003, the most notable increase had been on fuel cost due to the rise in oil price over the period. In contrast, salary costs had gone down considerably which could be resulted from some structural changes in the trade and the way goods vehicles were operated.

E.2.15 Generally, there were marginal increases of 4 - 5% in the annual total expenses from 2003 to 2011 in nominal terms except van-type LGV. The annual total expenses incurred by van-type LGVs had dropped by 14% over that period and it could be due to some major changes in their usage (e.g. increased non-commercial use) and operation. In real terms, the operating costs for all vehicle types had dropped between 11% and 27% from 2003 to 2011.

E.2.16 As a result of this drop in annual expenses in real terms, coupled with the increase in average daily distance travelled as earlier mentioned, the average goods vehicle operating cost (inclusive of all costs including capital cost of vehicle) per kilometre travelled had dropped from HK$13.1/km (in 2011 prices) in 2003 to HK$9.2/km (in 2011 prices) in 2011, a notable 30% reduction in real terms among all goods vehicles.

E.3 Characteristics of Goods Vehicle Trips Made on a Weekday

Total Goods Vehicle Trips Made

E.3.1 It is estimated that a total of 785,500 goods vehicle trips were made within the HKSAR territory (including trips from a place within the HKSAR to any Hong Kong boundary control point and vice versa) on a normal weekday, decreased from the 833,000 daily trips in 2003. 27% of these daily goods vehicle trips were SG-related, i.e. to and/or from the SG sites.

E.3.2 The distribution of goods vehicle trips by vehicle type is shown in **Figure E.3**. Van-type LGV trips constituted the largest proportion (40%) of the daily goods vehicle trips.

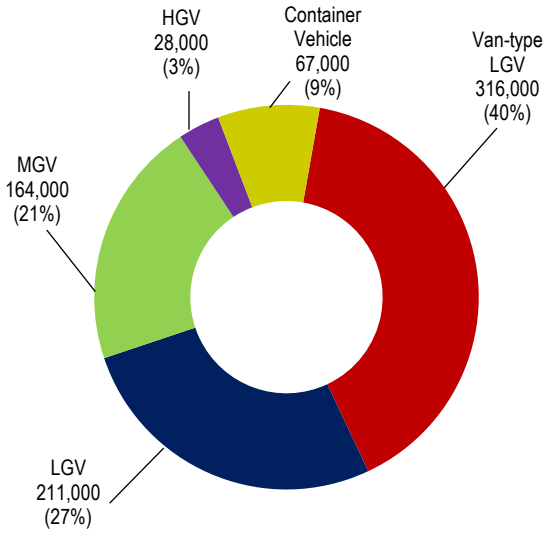


Figure E.3 : Daily Goods Vehicle Trips by Vehicle Type

E.3.3 **Table E.7** compares the 2003 and 2011 goods vehicle trip rates[2] by vehicle type. The overall goods vehicle trip rate on a weekday had decreased slightly from 7.40 trips/vehicle in 2003 to 7.02 trips/vehicle in 2011. In comparison, the decrease in trip rate was more noticeable for van-type LGV and MGV for which close to 10% reductions were observed.

Table E.7 Weekday Daily Goods Vehicle Trip Rates by Vehicle Type in 2003 and 2011

Vehicle Type	Trip Rate (Daily Trips Per Vehicle)	
	2003	2011
Van-type LGV	7.90	7.23
LGV	7.80	7.73
MGV	6.70	6.03
HGV	7.90	7.95
Container Vehicle	6.40	6.48
Overall	**7.40**	**7.02**

Trip-making Time

E.3.4 As compared to person trips, goods vehicle trips stayed generally flat from 9:00 a.m. to 6:00 p.m. with a trough around lunch hour at 1:00 – 2:00 p.m. The peak periods occurred during 10:00 a.m. – 12:00 noon in the morning and 2:00 – 4:00 p.m. in the afternoon. These two peak periods accounted for 20% and 17% of the daily trip total respectively. This pattern was generally consistent across all types of goods vehicles, and was also similar to that in GVTCS 2003.

Trip Journey Time

E.3.5 The survey results revealed that close to 74% of the trips took 30 minutes or less to complete; 22% took more than 30 minutes to an hour; and about 4% took more than an hour. The mean journey time was 27 minutes.

E.3.6 Among different vehicle types, trips made by van-type LGVs and LGVs were generally shorter with a mean journey time of 24 to 25 minutes. HGV and container vehicle trips were longest with mean journey times of 34 and 41 minutes respectively.

[2] Goods vehicle trip rate refers to the average number of trips made on a daily basis per vehicle.

Trip Movements

E.3.7 The estimated daily goods vehicle trip movements on a normal weekday are summarised in **Table E.8** by main geographical region.

Table E.8 Major Trip Movements

To/ From	HKI	Kowloon	NT	BCP	Total
HKI	77,500	21,500	10,200	100	**109,300**
Kowloon	19,400	166,900	72,200	500	**259,000**
NT	8,500	71,000	317,200	10,800	**407,500**
BCP	100	800	8,700	N/A	**9,700**
Total	**105,500**	**260,300**	**408,400**	**11,400**	**785,500**

Legend: "HKI" denotes "Hong Kong Island"; "NT" denotes "New Territories"; "BCP" denotes "Boundary Control Point"

E.3.8 Intra-NT movement accounted for the largest proportion (40%) of the total number of daily trips. Analysis of the goods vehicle trip origin and destination locations indicates that the most common origin/destination district for goods vehicle trips was Kwai Chung which accounted for about 9% of both origin and destination trip ends.

E.3.9 This was attributed to the frequent activities of container vehicles and other heavy goods vehicles to serve the Container Terminal and other backup facilities located in the district. For van-type LGVs and LGVs, their trip ends were more dispersed.

E.3.10 Cross-boundary trips constituted less than 3% of the daily goods vehicle trip total. This was decreased from the close to 4% in the GVTCS 2003.

Loading Conditions

E.3.11 The loading conditions of goods vehicles by vehicle type are illustrated in **Figure E.4**. HGV trips showed the largest proportion (36%) of "full" loading (defined as carrying more than 80% of the loading capacity), while Container vehicle trips showed the largest proportion (66%) of "empty" loaded (defined as carrying less than 10% of the loading capacity).

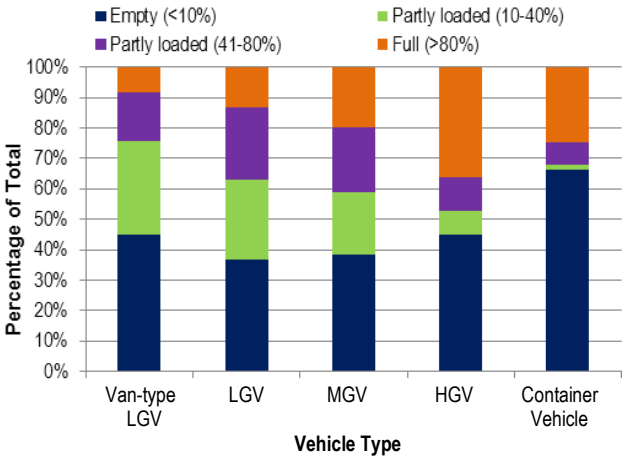


Figure E.4 : Loading Conditions by Vehicle Type

Usage of Toll Roads/Tunnels

E.3.12 Most (76%) of the trips reported did not involve the use of any toll road/tunnel. Among the various tunnels, Cross Harbour Tunnel was the most frequently used. 5% of the surveyed trips included the use of Cross Harbour Tunnel, followed by Tseung Kwan O Tunnel (3%), Lion Rock Tunnel (3%) and Shing Mun Tunnel (3%). The remaining (10%) of the surveyed trips used other toll roads/tunnels.

E.4 Next Step

E.4.1 One of the prime objectives of the GVTCS 2011 is to collect up-to-date goods vehicle trip characteristics data and information to facilitate the continuous enhancement of the CTS model. The data collected will be used in the re-calibration and enhancement of the goods vehicle trip model within the CTS model as the next stage of work for the Consultancy Study.

1. INTRODUCTION

1.1 Background

1.1.1 The last Survey on Goods Vehicle Trip Characteristics (GVTCS 2003) was conducted in 2003 and completed in 2004. GVTCS 2003 collected information on the trip-making characteristics of goods vehicles in Hong Kong, and archived a database for updating the goods vehicle trip model within the Government's comprehensive transport model, which has been widely applied in various major territorial and sub-regional transport studies in Hong Kong.

1.1.2 In order to keep the transport model up-to-date, the Transport Department commissioned Arup (the Consultant) in July 2010 to undertake the "Comprehensive Transport Study Model Enhancement – Feasibility Study" (the Consultancy Study) under Agreement No. CE 67/2009. The Consultancy Study defined the scope and data requirements of the GVTCS 2011 for transport planning and modelling purposes; and will apply the collected data in subsequent enhancement of the Comprehensive Transport Study (CTS) model for better simulation of the latest patterns of goods vehicle movements in Hong Kong.

1.1.3 The planning, organisation and execution of the GVTCS 2011 surveys proposed by the Consultancy Study were undertaken under a separate service contract "TD 3/2011 – Provision of Services for Conducting the Survey on Goods Vehicle Trip Characteristics 2011" (the Service Contract). The Service Contract was awarded to MVA Hong Kong Ltd (MVA) in July 2011.

1.2 Study Objectives

1.2.1 The GVTCS 2011 formed an integral part of the Consultancy Study. Its aim was to collect up-to-date goods vehicle operation and trip-making characteristics data for enhancing the Goods Vehicle Sub-model of the CTS Model. Furthermore, the collected data was archived into a database which allows transport planners to better understand the characteristics of goods vehicle movements and to facilitate future planning for goods vehicles traffic.

1.2.2 Specific objectives of the GVTCS 2011 are:

- To define the scope and data requirements of the surveys;
- To collect up-to-date trip characteristics data of goods vehicles for modelling and planning purposes;
- To archive a database from the survey data and develop procedures for analysing the data;
- To assess changes in goods vehicle trip characteristics against the previous GVTCS or relevant surveys; and
- To review and make recommendations on the approach to conduct future survey updates.

1.2.3 GVTCS 2011 did not include any re-calibration and enhancement work of the CTS Model. Such work is being carried out as the next phase of the Consultancy Study.

1.3 Study Approach and Process

1.3.1 **Figure 1.1** illustrates the GVTCS 2011 process. The work was carried out in 4 phases involving the following key tasks.

Design Phase

- Review of CTS Goods Vehicle Trip Model parameters
- Establishment of scope and data requirements of surveys
- Outline design of interview surveys

Tender Phase

- Preparation of tender documents for the Service Contract
- Determination of evaluation criteria and marking scheme
- Tender assessment and recommendation

Data Collection Phase

(Primary responsibility of the Service Contractor, under the supervision and management of the Consultant as required by the Consultancy Study)

- Establishment of questionnaires and fieldwork procedures
- Pilot and main survey fieldwork
- Quality control
- Data processing
- Data checking and verification

Data Analysis / Reporting Phase

- Trip data expansion
- Data analysis and tabulations
- Recommendations on future survey update strategies
- Archive goods vehicle trip database, handover and training

1.4 Structure of this Report

1.4.1 Following this introductory section, the remainder of this Report is structured as follows:

- Section 2 summarises the survey design, fieldwork implementation, data processing and database development for the GVTCS 2011.
- Section 3 presents some key findings with respect to the profile of the goods vehicle owners and the usage of goods vehicles.
- Section 4 details the characteristics and patterns of the goods vehicle trips made within the Hong Kong Special Administrative Region (HKSAR) on a weekday.
- Section 5 summarises some key results obtained from the goods vehicle trip generation and composition surveys.
- Section 6 provides a summary comparison of the GVTCS 2011 results with the previous surveys.
- Section 7 discusses the next step with respect to the use of the GVTCS 2011 data.

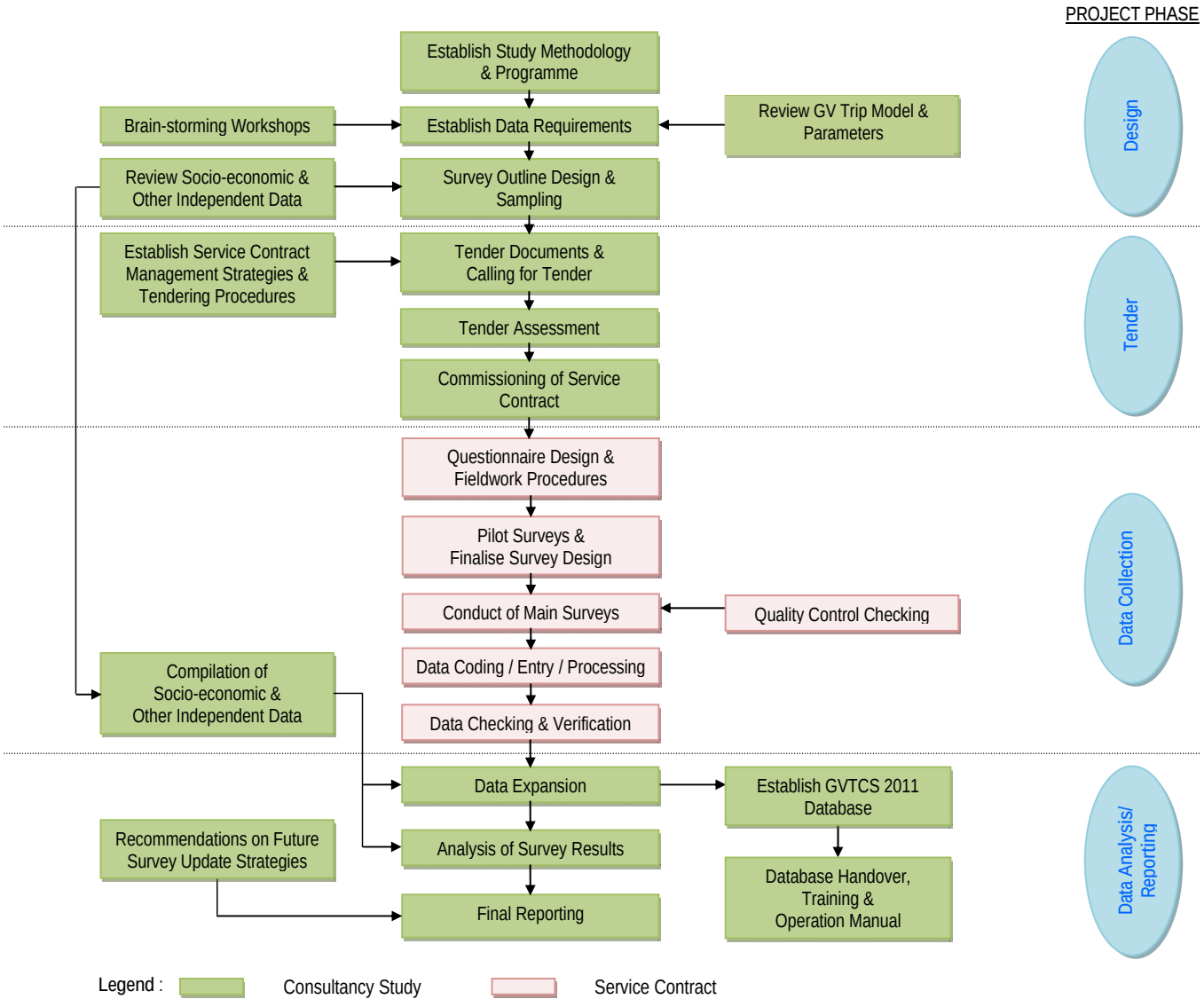


Figure 1.1 : Study Process

2. SURVEY DATA COLLECTION AND ASSIMILATION

2.1 Survey Scope and Data Requirements

2.1.1 The GVTCS 2011 surveys were devised to collect updated and comprehensive goods vehicle trip characteristics data, firstly for the re-calibration and enhancement of the Goods Vehicle Trip Model within the CTS Model, and secondly for better appreciation of the goods vehicle operating and trip characteristics to facilitate future transport planning. The key of the survey design process was to establish what data had to be collected in the GVTCS 2011 in order to fulfil the above purposes.

2.1.2 An in-depth review of the existing CTS Goods Vehicle Trip Model was undertaken to determine the type of data necessary for derivation of relevant parameters. In addition to the data collected in the previous GVTCS 2003, some extra types of data were proposed to obtain additional information for enhancement of the model capability, or to provide more insights into specific goods vehicle operation and trip characteristics.

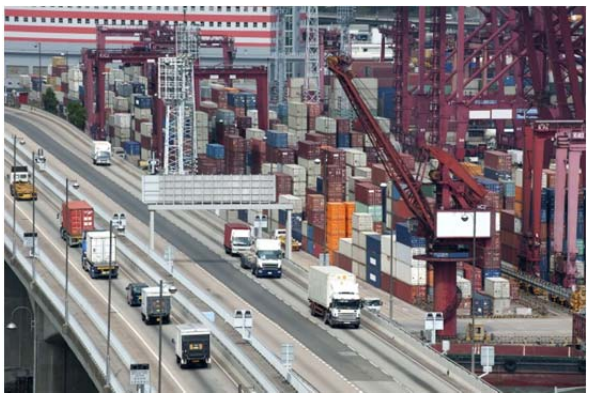
2.1.3 In addition, a series of brain-storming workshops were conducted with academics and representatives from various government bureaux/departments to identify potential improvements to the existing CTS Model as well as to discuss their specific data needs.

2.1.4 Resulting from the above review, the scope of the GVTCS 2011 was defined to cover the following five surveys:

- Goods Vehicle Owner and Driver Interview Survey (GVIS)
- Special Generator Interview Survey (SGIS)
- Goods Vehicle Trip Generation Survey (GVTGS)
- Goods Vehicle Composition Survey (GVCS)
- Vehicle Journey Time Survey (VJTS)

2.1.5 Detailed information on the goods vehicle trips made within the HKSAR on a normal weekday, which was the mainstay of the Goods Vehicle Trip Model development, were collected through the GVIS and SGIS. Besides the trip information, general information on the owner and vehicle characteristics as well as specific information related to vehicle operating costs were also collected through interviews with good vehicle owners in the GVIS.

2.1.6 Moreover, other supplementary information related to goods vehicles or their traffic movements were collected through the other surveys (GVTGS, GVCS and VJTS) either to provide additional information for general transport planning purposes or to provide independent data for subsequent GVTCS data processing or model validation purposes.



GVTCS 2011 was conducted to collect up-to-date goods vehicle trip characteristics data

2.2 Design of Surveys

Goods Vehicle Owner and Driver Interview Survey (GVIS)

2.2.1 This formed a major part of the GVTCS 2011 data collection and provided essential information on the trip patterns of goods vehicles operating within the HKSAR. The survey covered goods vehicles licensed with the Transport Department by random sampling, as well as goods vehicles owned by the Hong Kong Government. Goods vehicles were classified into five vehicle types as shown in **Table 2.1** and was consistent with the types of vehicles adopted in the transport model setup.

Table 2.1 Goods Vehicle Classifications

Vehicle Type	Description	Permitted Gross Vehicle Weight (tonnes)
Van-type LGV#	• LGV constructed with a fully enclosed body which is an integral part of the vehicle	≤ 5.5
Light Goods Vehicle (LGV)*	• Usually shorter than 6m • 2-axled light goods vehicle other than van-type LGV	≤ 5.5
Medium Goods Vehicle (MGV)*	• Usually longer than 6m • 2-axled with double rear tyres or 3-axled	> 5.5 to 24
Heavy Goods Vehicle (HGV)*	• Usually longer than 7m • Usually 4-axled	> 24 to 38
Container Vehicle#	• All articulated vehicles	≤ 44

* *Cap 374 - Road Traffic Ordinance*
\# *Cap 374A - Road Traffic (Construction and Maintenance of Vehicles) Regulations*

2.2.2 A random sample of goods vehicles was selected from the goods vehicle licensing records in the Vehicles and Drivers Licensing Integrated Data System (VALID) database maintained by the Transport Department (TD) as at September 2011. The licensed goods vehicles were stratified by vehicle type and geographical area of the vehicle registration address and then sampled by disproportionate stratified random sampling with an aim to achieve a minimum sample size for each stratum.

2.2.3 With the assistance of TD, a list of government departments which own goods vehicles was complied. The list included the numbers and types of goods vehicles owned by each of these departments, from which a sample of government goods vehicles was selected for survey.

2.2.4 Both the owners and drivers of the sampled goods vehicles were approached for face-to-face interviews. In general, information related to the business nature, usage, overnight parking location and operating costs of the sampled vehicles were obtained from the vehicle owners. Also, details of all the trips made using the sampled goods vehicles on a reference weekday (defined as the previous or last operating weekday within the two weeks before interview which was not a public holiday) including locations and land use types at trip origin and destination, trip purpose, trip starting and ending time, parking at trip ends, loading condition, number of delivery workers in the vehicle, use of toll facilities etc., were collected from the goods vehicle drivers.

Special Generator Interview Survey (SGIS)

2.2.5 Special generators (SGs) refer to sites as shown in **Figure A.1** in the **Appendix** that generate significant goods vehicle movements with specific trip characteristics. The main purpose of the SGIS was to supplement the GVIS by collecting trip generation and characteristics data on goods vehicle movements to and from these SGs; the GVIS alone, as a sample survey, would not be able to capture sufficient number of SG trip records for effective analysis.

2.2.6 For the purpose of this survey, the following types of SGs were defined:

- Kwai Tsing Container Terminal
- River Trade Terminal/ Mid-stream Site/ Private Wharf
- Public Cargo Working Area
- Wholesale Market
- Air Cargo Terminal
- Freight Forwarding Centre/ Logistic Centre/ Warehouse
- Open Storage and Port Back-up Facility
- Landfill Site and Barging Point
- Hospital

2.2.7 The frames for each SG type were compiled by the Consultant based on a combination of independent sources. A total of 80 sites were then randomly selected from individual frames and then verified by the Contractor through site visits before conducting the surveys.

2.2.8 For each surveyed site, two kinds of surveys were conducted to obtain data for subsequent derivation of goods vehicle trip matrices by vehicle type to and from the SGs:

(i) Intercept interviews with the goods vehicle drivers for collecting information on the trip characteristics (such as location and land use type at the other end of the trip, trip purpose, arrival/departure time at site, loading condition, number of delivery workers in the vehicle, use of toll facilities) of goods vehicle entering or leaving the site;

(ii) Classified vehicle counts at the entrances/ exits for collecting goods vehicle flows by vehicle type into and out of the site to expand the sample data obtained from interviews in (i) above.

2.2.9 Surveys were conducted at each selected SG site during their operating hours for at least two weekdays (excluding public holidays).

Goods Vehicle Trip Generation Survey (GVTGS)

2.2.10 The survey was carried out to establish trip generation of different types of goods vehicles generated by different land use developments for various time periods on a normal weekday.

2.2.11 The sampling frame was compiled based on a combination of information from the Education Bureau (for educational establishments), Rating and Valuation Department (domestic and non-domestic developments) and other independent sources. A total of 219 sites stratified by land use type and geographical sector were randomly selected from the frame and then verified by the Contractor through site visits to confirm the practicality of conducting the survey with a view to capturing all goods vehicle traffic generated by the subject sites including the associated on-street loading/ unloading activities.

2.2.12 Classified traffic counts were conducted at each surveyed site for 24 hours (or such duration when traffic could be generated by the site) of at least one weekday, excluding public holiday. All goods vehicles generated by the surveyed sites, including those loading/ unloading on-street, were counted and recorded at 15-minute intervals by direction (arrival/ departure).

Goods Vehicle Composition Survey (GVCS)

2.2.13 The survey was to collect traffic flow data for estimating the composition of goods vehicle traffic by vehicle type on major roads in Hong Kong at different times of a normal weekday. The data was used as basis for validating the goods vehicle matrices developed from the GVIS and SGIS data.

2.2.14 The survey was conducted at all the 104 traffic count stations forming the cordons and screenlines defined in the Annual Traffic Census (including the four cordons: Hong Kong External, Hong Kong Internal, Kowloon External and Tsing Yi External; and eleven screenlines: A-A, C-C, F-F, G-G, H-H, I-I, K-K, R-R, S-S, T-T, Y-Y) and the screenline formed by the three cross-harbour tunnels, as well as three additional cordons defined by the Study for future model validation purpose, namely: Shatin External, Ma On Shan External and Tseung Kwan O External. Refer to **Figures A2-A4** in the Appendix for the locations of these screenlines/cordons.

2.2.15 At each count station, classified goods vehicle counts were undertaken from 7:00 a.m. to 9:00 p.m. for at least one normal weekday, excluding public holiday. Goods vehicle traffic flows distinguished by the five vehicle types in Table 2.1 were recorded at 15-minute intervals by direction.

Vehicle Journey Time Survey (VJTS)

2.2.16 The purpose of this survey was to obtain supplementary information on the average journey time taken by vehicles to traverse selected road links for reviewing the average operating speed of traffic in the road network.

2.2.17 Besides reference made to TD's "Car Journey Time Survey for Monitoring Traffic Congestion 2010 and 2011", the vehicle journey time survey was conducted for 29 routes.

2.2.18 The survey was conducted during the morning (7:15 – 9:30 a.m.) and evening (5:00 – 7:30 p.m.) peak periods.

2.2.19 The Moving Car (Modified) Method as stipulated in the Transport Planning and Design Manual was adopted for this survey, which was also consistent with TD's "Car Journey Time Survey for Monitoring Traffic Congestion 2010 and 2011". At least four runs were conducted for each survey route for each peak period concerned. In case of incidents which might affect the journey time, including traffic accident before or during the survey, roadworks, temporary traffic diversion and inclement weather, additional re-runs were conducted to replace the affected survey results. In case of roadworks and temporary traffic diversion, additional re-runs for replacement were conducted after completion of the concerned roadworks.

2.3 Survey Fieldwork Implementation

2.3.1 The operation, procedures and questionnaires including the design of data record forms of the respective surveys were tested through pilot surveys, with necessary refinements made for final adoption in the main fieldwork. The surveys were suspended during the Christmas/New Year holiday period to avoid skewed effect of travel patterns during holidays.

2.3.2 The fieldwork period and enumeration of the respective surveys are summarised in **Table 2.2**.

Table 2.2 Fieldwork Period and Enumeration

Survey	Fieldwork Period	Enumeration
Goods Vehicle Owner and Driver Interview Survey (GVIS) *(Response Rate: 70%)*	10 Oct 2011 – 22 Feb 2012	4,470 Non-government vehicles + 121 government vehicles
Special Generator Interview Survey (SGIS)	27 Oct 2011 – 13 Jan 2012	80 sites involving 27,220 trip records (2-way)
Goods Vehicle Trip Generation Survey (GVTGS)	6 Oct 2011 – 20 Dec 2011*	219 sites
Goods Vehicle Composition Survey (GVCS)	6 Oct 2011 – 20 Dec 2011#	104 stations
Vehicle Journey Time Survey (VJTS)	6 Oct 2011 – 16 Jan 2012	29 routes

* Supplementary surveys were conducted at two sites on 17 April 2012.
Re-surveys were conducted at three sites on 7-8 February 2012.

2.3.3 The number of surveyed sites and trip records enumerated by the SGIS broken down by the type of special generator are given in **Table 2.3**.

Table 2.3 SGIS Enumeration by Type of Special Generator

SG Type	Number of Surveyed Sites	Number of Trip Records (2-way) Enumerated
Kwai Tsing Container Terminal	4	2,170
River Trade Terminal/ Mid-stream Site/ Private Wharf	4	2,300
Public Cargo Working Area	3	2,130
Wholesale Market	5	960
Air Cargo Terminal	2	2,540
Freight Forwarding Centre/ Logistic Centre/ Warehouse	30	9,500
Open Storage and Port Back-up Facility	17	2,080
Landfill Site and Barging Point	5	4,120
Hospital	10	1,420
Total	**80**	**27,220**



Wholesale market has been included as a new category of Special Generators in the 2011 SGIS

2.3.4 Throughout the fieldwork period, quality control measures were strictly applied to ensure that data of the highest quality were collected. These were implemented by an independent quality control team of the Contractor and also by the Consultant. Apart from the training and day-to-day supervision of the fieldwork, 15% of the completed GVIS questionnaires were randomly selected by independent checkers for back-checking to verify that interviews with the goods vehicle owners and drivers were made and key information collected was correct. In addition, random on-site checks were made for on the work and quality of the completed questionnaires/data record forms for the other surveys.

2.4 Data Processing and Expansion

2.4.1 All the data collected in the interview surveys were edited, coded and entered in computer format by the Contractor. The addresses given for fields involving locations such as trip origin, destination and interchange locations (between transport modes taken) were coded to the Tertiary Planning Unit Street Block level. To minimise the risk of incorrect data entry, a double data entry system was adopted, i.e. all data were keypunched twice into the computer by different staff for verification.

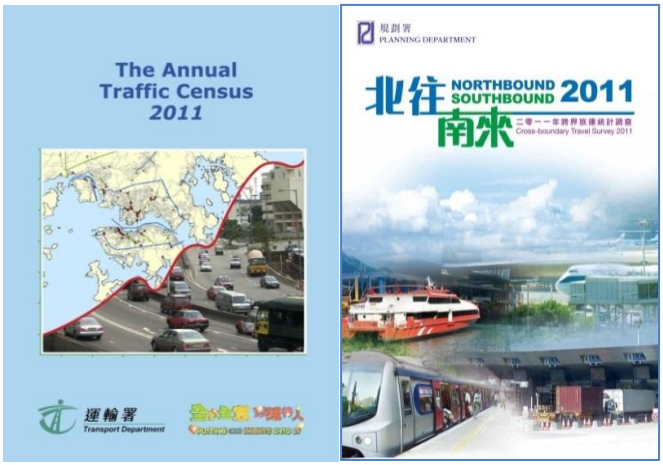
2.4.2 The input data went through a series of validity checks using the computer to identify any duplicated records, omitted items, out-of-range values and inconsistencies between different data items, for subsequent verification and cleaning.

2.4.3 The cleaned data collected from the GVIS and SGIS were then expanded to represent the population based on a 2-stage process. The first stage involved direct expansion of the GVIS results according to the sampling fractions by stratum to match the goods vehicle fleet size control totals, while the SGIS data for each surveyed site was initially expanded to match the observed traffic counts at individual sites conducted in tandem with the interviews.

2.4.4 As the SGIS was only conducted at a sample of selected sites, the data collected at the surveyed sites, after initial direct expansion to match observed counts, had to be further expanded to the territory-wide SG totals based on relevant parameters such as number of sites, throughput, site or floor area, as appropriate.

2.4.5 Following the above, initial goods vehicle trip matrices for SG-related and non-SG-related trips were derived from the SGIS and GVIS results respectively. They were then compared against independent observed traffic flows across screenlines/cordons and adjusted accordingly by using matrix estimation technique. Matrix adjustments in this second stage of expansion were necessary because, irrespective of the survey design, some under-reporting of trips by the GVIS respondents would be inevitable. For the SGIS trip data, the adjustments mainly accounted for any discrepancy arising from the application of the surveyed SG site characteristics to the SG population totals.

2.4.6 In this stage of expansion process, the GVCS traffic counts were adopted as the control goods vehicle flows. Reference has also been made to the Annual Traffic Census daily traffic profiles. For cross-boundary goods vehicle movements, PlanD's Cross-boundary Travel Survey 2011 results were applied as controls.



Independent observed traffic statistics from various sources including the GVCS results, Cross-boundary Travel Survey and Annual Traffic Census were adopted for GVTCS trip data expansion

2.5 GVTCS 2011 Database

2.5.1 The finally expanded data for the GVIS and SGIS were archived into a comprehensive goods vehicle trip characteristics database which comprises key data items on the owner, vehicle and trip information given in **Table 2.4.**

2.5.2 The GVTCS 2011 database will be adopted as the basis for the re-calibration and updating of the CTS Goods Vehicle Trip Model. In addition, the information in the database will provide useful reference for future goods vehicle transport planning.

Table 2.4 Key Data Items in the GVTCS 2011 Database

Data Type		Key Data Items
GVIS	Owner	• Industry Engaged in * • Goods Vehicle Fleet Size Owned • Total Number of Employees • Number of Drivers and Delivery Staff Employed
	Vehicle	• Type of Goods Vehicle • Main Purpose of Vehicle Usage • Usual Location for Overnight Parking (or When Vehicle Not in Operation) • Usual Type of Parking Space and Average Fee Paid for Overnight Parking (or When Vehicle Not in Operation) • Age of Vehicle, Expected Life and Price of Purchase • Total Working Hours of Drivers/ Delivery Staff Deployed Per Week • Expenses Incurred in Operating the Vehicle • Decision Maker and Any Criterion for Route Choice
	Trip	• Locations of Trip Origin/ Destination (in Tertiary Planning Unit Street Block) • Trip Starting and Ending Time, and Journey Time • Land Use at Trip Origin/ Destination • Type of Parking Space at Trip Origin/ Destination • Trip Purpose • Number of Persons in the Vehicle • Loading Condition • Control Point Used for Cross-boundary Trips • Use of Tunnels/ Tolled Roads
SGIS	Trip	• Type of Goods Vehicle • Direction (Inbound/ Outbound) • Time Arriving/ Departing Site • Location of the Other End of the Trip (in Tertiary Planning Unit Street Block) • Land Use at the Other End of the Trip • Trip Purpose • Control Point Used for Cross-boundary Trips • Use of Tunnels/ Tolled Roads • Loading Condition • Number of Persons in the Vehicle

* For non-government vehicles only

3. GOODS VEHICLE OWNER AND VEHICLE USAGE

3.1 Profile of Goods Vehicle Owners

3.1.1 Among the non-government goods vehicles, the largest proportion of their owners were engaged in the "Transportation, Storage, Postal and Courier Services" sector (55%) as shown in **Figure 3.1**. It was followed by "Construction" (15%) and "Wholesale, Retail and Import/Export Trade" (13%). These percentages were fairly consistent with the corresponding figures of 49%, 15% and 14% in GVTCS 2003.

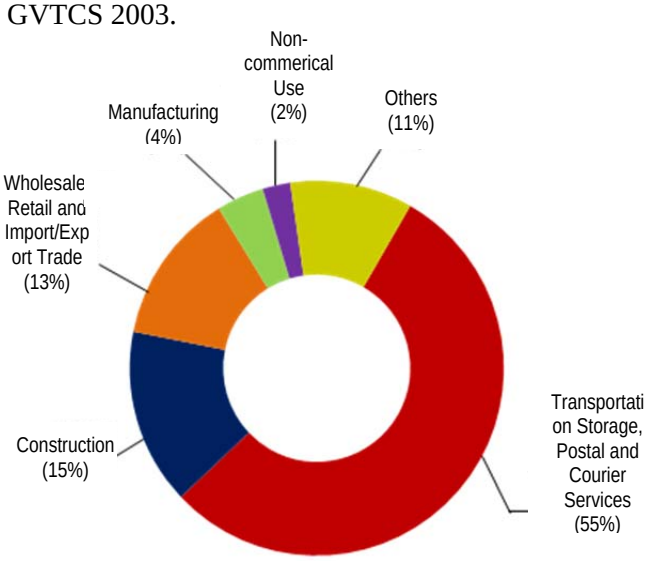


Figure 3.1 : Industry Goods Vehicle Owners Engaged In

3.1.2 "Manufacturing" industry accounted for only 4% of the goods vehicles, which dropped from 8% in 2003. About 2% of the goods vehicles were for non-commercial use. Most of them were van-type LGVs.

3.2 Goods Vehicle Usage

3.2.1 As at September 2011, the total goods vehicle fleet size in Hong Kong was 111,946 vehicles, comprising 110,660 goods vehicles licensed in TD's VALID database (non-government vehicles) and another 1,286 goods vehicles owned by the HKSAR government (government vehicles). This was about 1% less than the total fleet size of 113,380 goods vehicles as reported in GVTCS 2003.

3.2.2 The proportions of goods vehicles by vehicle type are illustrated in **Figure 3.2**. As shown, van-type LGVs accounted for the largest proportion (39%) of the goods vehicle fleet, followed by LGVs (25%) and MGVs (24%).

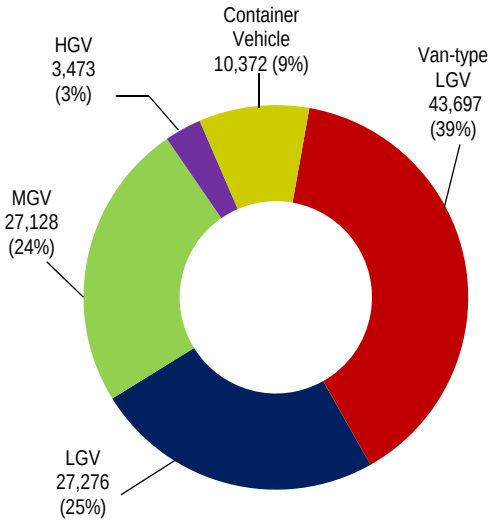


Figure 3.2 : Goods Vehicle Fleet by Vehicle Type

3.2.3 As compared to 2003, the proportion of van-type LGVs had increased from 34% to 39%. Corresponding reductions in proportions were observed for LGVs (from 27% in 2003) and container vehicles (from 12% in 2003).

3.2.4 About half (49%) of the non-government goods vehicles were mainly used for transporting goods of the respondents' own company. 36% mainly provided transport services directly to customers. "Transportation of employees of own company" and "personal transport by respondents or their family" each accounted for 5%, which comprised mostly van-type LGVs and to a lesser extent LGVs.

3.2.5 As for the government goods vehicles, most (68%) of them were mainly used for transporting goods of the respondents' own department. 17% were used mainly for transportation of operational staff of the respondents' own department, and 10% mainly for repair/ maintenance/ carrying out works/ cleaning.



The number of van-type LGVs had increased by 12% from 2003 to 2011

3.2.6 The average numbers of working hours of drivers and delivery staff deployed for each goods vehicle per week are given in **Table 3.1**. On average, 48 hours of drivers' and 39 hours of delivery staff's time were deployed in a week for each goods vehicle.

Table 3.1 Average Numbers of Working Hours of Goods Vehicle Drivers and Delivery Staff Per Week

Vehicle Type	Average Number of Working Hours (1) Per Week	
	Driver	Delivery Staff
Van-type LGV	45.0	34.7
LGV	46.8	42.7
MGV	50.0	43.0
HGV	51.8	39.0
Container Vehicle	54.7	13.0
Overall	**48.0**	**39.3**

Note: (1) All goods vehicles excluding non-commercial use.

3.2.7 As indicated from the above table, the number of drivers' working hours generally increased with the Gross Vehicle Weight (GVW) of the goods vehicle. On the other hand, the number of delivery staff's working hours was highest among MGVs and LGVs.

3.2.8 The number of working hours of drivers would be a proxy of the operating hours of the goods vehicle. On this basis, the average operating hours of the goods vehicles was estimated to be about 2,500 hours per year. This was slightly more than the 2,420 hours estimated in GVTCS 2003.

3.2.9 **Table 3.2** shows the average annual kilometrage travelled by different goods vehicle types, and compares them against the GVTCS 2003 results. In 2011, the overall average annual distance travelled by a goods vehicle including cross-boundary trips was 38,300 km. This was notably higher than the 31,800 km estimated in GVTCS 2003 for which confined to travel distances within HKSAR territory only. As a relatively large proportion of container vehicle trips involved cross-boundary movements, the average annual kilometrage travelled by container vehicles in 2011 was much higher than in 2003.

Table 3.2 Average Annual Kilometrage Travelled by Goods Vehicle Type in 2003 and 2011

Vehicle Type	Annual Kilometres Travelled (km) (1)	
	2003	2011
Van-type LGV	30,300	27,700
LGV	29,900	31,200
MGV	32,500	46,400
HGV	39,800	46,900
Container Vehicle	36,900	78,900
Overall	**31,800**	**38,300**

Note: (1) The GVTCS 2011 covered the distances travelled outside the HKSAR for cross-boundary trips while the GVTCS 2003 were confined to that within the HKSAR territory only.

3.2.10 Among different types of goods vehicles, the average annual kilometrage travelled generally increased with the size or GVW of the vehicle, starting from 27,700 km for van-type LGV to the highest 78,900 km for container vehicle.

3.3 Overnight Parking of Goods Vehicles

3.3.1 Goods vehicle owners were asked where their vehicles were parked overnight (or when the vehicles were not in operation) over the 6 months before enumeration. The results indicated that the majority (95%) of goods vehicles were parked within Hong Kong. Only 5% of goods vehicles were parked in the Mainland, as compared to 10% in 2003.

3.3.2 Comparing different vehicle types, a much higher percentage (22%) of container vehicles were parked overnight in the Mainland. It was followed by MGVs (11%). Other goods vehicle types generally had less than 5% parking

in the Mainland, with van-type LGV showing the lowest proportion (less than 0.5%).

3.3.3 When the goods vehicles were parked overnight in Hong Kong, the distribution of their most usual parking locations are set out in **Table A.1** in the Appendix by 26 broad districts commonly adopted for transport planning and modelling analysis (refer to **Figure A5** in the Appendix for definition of these districts). In general, 8% of them were parked on Hong Kong Island, 21% in Kowloon, and 71% in the New Territories (NT). These compared to 8%, 18% and 74% respectively in 2003.

3.3.4 Among the 26 broad districts, rural Northwest NT (13%) and Kwai Chung (11%) accounted for the largest proportions of overnight goods vehicle parking.



Goods vehicles parking on streets

3.3.5 The types of parking space used by goods vehicles when parked overnight in Hong Kong are summarised in **Table 3.3**. The vast majority (90%) were parked off-street, with monthly rental space (57%) being the most common type of parking, followed by designated rent-free space (18%) and self-owned space (8%).

Table 3.3 Overnight Parking Arrangement in Hong Kong

Type of Parking Space		Proportion of Goods Vehicles
Off-street	Self-owned space	8%
	Rent-free space (designated)	18%
	Monthly rental space	57%
	Hourly/daily rental space	3%
	Designated loading/unloading space	*
	Non-designated	4%
On-street	Metered	4%
	Designated non-metered	*
	Non-designated	5%
Total		**100%**

* Less than 0.5%

3.3.6 The average monthly costs paid for parking goods vehicles (excluding government vehicles) overnight in Hong Kong are illustrated in **Figure 3.3** by vehicle type. The average parking cost ranged from the lowest $910/month for HGVs to the highest $1,630/month for container vehicles. Average overnight parking cost was $1,160/month. The lower parking cost for HGVs could be attributed to the larger proportion of HGVs parked at non-designated parking spaces (both on-street and off-street) as compared to other vehicle types.

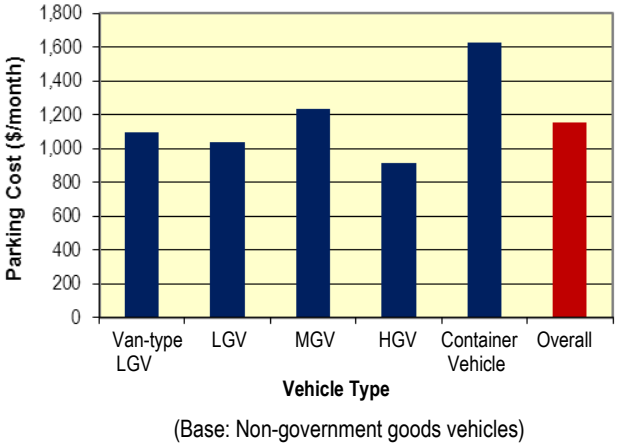


Figure 3.3 : Average Overnight Parking Costs Paid for Goods Vehicles

3.3.7 All government goods vehicles were parked at designated off-street rent-free spaces, therefore they did not incur any parking cost.

3.4 Costs for Operating Goods Vehicles

3.4.1 Detailed information was obtained from goods vehicle owners about the average expenses incurred in operating their vehicles. **Table 3.4** summarises the annual expenses, itemised by type of expenses and vehicle type.

3.4.2 The average total annual operational cost (excluding capital cost of vehicle) incurred by a goods vehicle ranged from some $238,200 for a van-type LGV to $471,600 for a container vehicle. As can be expected, the operating cost generally increased with the size or GVW of the vehicle.

3.4.3 As compared to the GVTCS 2003 results, there were marginal increases (4% - 5%) in the annual total expenses from 2003 to 2011 in nominal terms for all vehicle types except van-type LGV. Such marginal increases in the total expenses (except for van-type LGVs) however fell behind inflation over the period. There was a 14% reduction in annual total expenses incurred by van-type LGVs and it could be due to some major changes in their usage (e.g. increased non-commercial use) and operation.

3.4.4 In real terms, the operating costs for all vehicle types had dropped by 11% - 27% between 2003 and 2011.

3.4.5 Amongst the operating expenses of goods vehicles, salary costs constituted the highest proportion of the total, varying between 40% and 60% for different vehicle types. Fuel cost came second (17% - 41%), followed by tyre replacement/ repair/ maintenance costs for MGV/ HGV/ Container Vehicle or parking cost for van-type LGV/ LGV. As compared to 2003, the most notable increase had been on fuel cost due to the rise in oil price over the period. In contrast, salary costs had gone down considerably which could be resulted from some structural changes in the trade, operation of goods vehicle and worker population.

Table 3.4 Annual Expenses Incurred in Operating Goods Vehicles

Expenses	GVan	LGV	MGV	HGV	CV
Parking Fee	20,370 (15,000)	19,770 (16,300)	23,270 (18,800)	15,660 (17,700)	28,300 (19,000)
Fuel Cost	40,510 (30,500)	60,390 (35,900)	112,270 (73,600)	171,070 (99,000)	194,070 (115,300)
Tyre Replacement/ Repair/ Maintenance Cost	10,810 (9,900)	16,680 (11,100)	30,130 (21,800)	51,710 (32,200)	44,550 (26,900)
Insurance Premium	5,390 (5,100)	6,150 (5,400)	9,070 (9,300)	13,080 (13,600)	13,990 (9,200)
Licence Fee	4,160 (4,300)	2,390 (2,400)	$4,670 (4,700)	4,690 (4,700)	4,690 (4,700)
Sub-total Operational Cost	**81,240 (64,800)**	**105,380 (71,100)**	**179,410 (128,200)**	**256,210 (167,200)**	**285,600 (175,100)**
Salary Costs	***157,000 (211,800)***	***178,500 (202,700)***	***208,260 (243,900)***	***199,270 (264,800)***	***185,940 (273,900)***
Annualised Vehicle Cost	***20,160*** **(18,700)**	***24,360*** **(25,200)**	***41,790*** **(42,400)**	***64,230*** **(63,000)**	***46,430*** **(62,700)**

Legend: "GVan" denotes "Van-type LGV"; CV denotes "Container Vehicle"

Note: 2003 figures are shown in ( ).

Monetary values above are expressed in nominal terms (i.e. money-of-the-day). Between 2003 and 2011, the Composite Consumer Price Index has increased by 18.2%.

4. CHARACTERISTICS OF GOODS VEHICLE TRIPS MADE ON A WEEKDAY

4.1 Total Goods Vehicle Trips Made

4.1.1 It is estimated from the expanded GVIS and SGIS results that a total of 785,500 goods vehicle trips were made within the HKSAR territory (including trips from a place within the HKSAR to any Hong Kong boundary control point and vice versa) on a normal weekday (Mondays to Fridays except public holidays). This had decreased from the 833,000 daily trips in 2003, representing a 6% reduction over the 8-year period.

4.1.2 **Figure 4.1** illustrates the distribution of goods vehicle trips by vehicle type. Van-type LGV trips constituted the largest proportion (40%) of the daily goods vehicle trips. It was followed by LGV (27%) and MGV (21%). HGV and container vehicle accounted for relatively small proportions of the daily goods vehicle trips at 3% and 9% respectively.

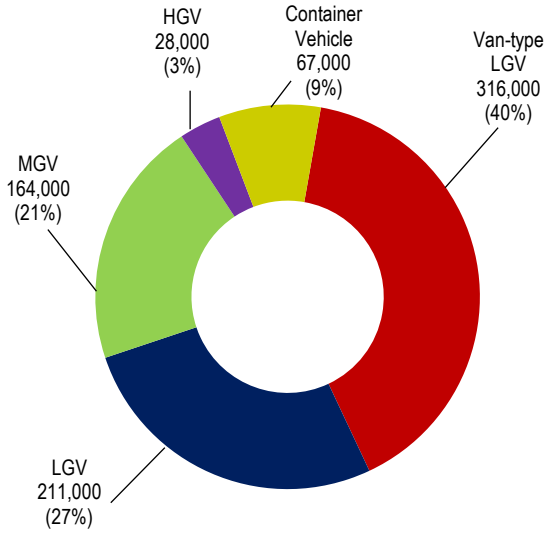


Figure 4.1 : Daily Goods Vehicle Trips by Vehicle Type

4.1.3 **Table 4.1** provides further breakdown of the daily goods vehicle trips into SG- and non-SG-related[1] trips by vehicle type.

Table 4.1 Daily SG-related and Non-SG-related Goods Vehicle Trips by Vehicle Type

Vehicle Type	Number of Trips		Proportion of SG-related
	SG-related	Non-SG-related	
Van-type LGV	32,600	283,300	10%
LGV	46,800	164,200	22%
MGV	54,000	109,700	33%
HGV	12,100	15,500	44%
Container Vehicle	64,100	3,100	95%

Note: Special generator (SG) includes those land uses defined for the purpose of the SGIS as well as boundary control points.

4.1.4 Overall speaking, 27% of the daily goods vehicle trips were SG-related, i.e. to and/or from the SG sites. The proportion of SG-related trips varied substantially across vehicle types. The proportion was the smallest (10%) among van-type LGV trips but was predominant (95%) for container vehicle trips.



A predominant proportion of the container vehicle trips started from or ended at the special generator sites.

4.1.5 Goods vehicle trip rates which refer to the average number of trips made on a daily basis per vehicle are presented in **Table 4.2** by vehicle type. They are compared with the 2003 trip rates in the table.

[1] SG-related trips refer to trips with one or both ends of the trip originating from or destined to any of the special generator land use types mentioned in Para. 2.2.6. Although not covered by the SGIS surveys, boundary control points are commonly considered as special generators of goods vehicle traffic. Therefore, cross-boundary trips are also classified as SG-related trips.

Table 4.2 Weekday Daily Goods Vehicle Trip Rates by Vehicle Type in 2003 and 2011

Vehicle Type	Trip Rate (Daily Trips Per Vehicle)	
	2003	2011
Van-type LGV	7.90	7.23
LGV	7.80	7.73
MGV	6.70	6.03
HGV	7.90	7.95
Container Vehicle	6.40	6.48
Overall	**7.40**	**7.02**

4.1.6 The overall weekday daily goods vehicle trip rate decreased slightly from 7.40 trips/vehicle in 2003 to 7.02 trips/vehicle in 2011. In comparison, the decrease in trip rate was more noticeable for van-type LGV and MGV for which close to 10% reductions were observed. Trip rates for the other vehicle types stayed relatively unchanged over the period.

4.2 Trip-making Time

4.2.1 **Figure 4.2** illustrates the profiles of goods vehicle trips made against different times of the day for various vehicle types. Trip-making time is defined as mid-point between trip arrival and departure times for this purpose.

4.2.2 Unlike person commuting trips, goods vehicle trips were more concentrated in the mid-day. They stayed generally flat from 9:00 a.m. to 6:00 p.m. with a trough around lunch hour at 1:00 – 2:00 p.m. The peak periods occurred during 10:00 a.m. – 12:00 noon in the morning and 2:00 – 4:00 p.m. in the afternoon. These two peak periods accounted for 20% and 17% of the daily trip total respectively. This pattern was generally consistent across all types of goods vehicles, and was also very similar to that in GVTCS 2003.

4.2.3 The profiles for trip-making time for SG-related and non-SG-related trips were similar, except that the SG-related trips tended to start earlier in the day and showed a less prominent afternoon peak.

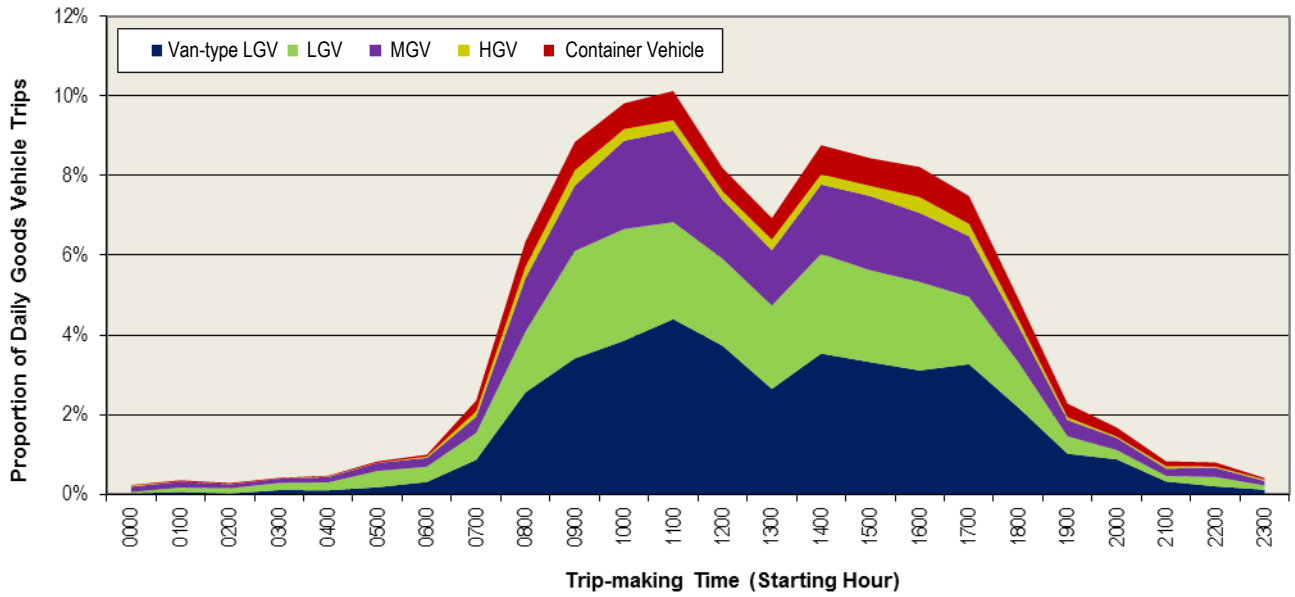


Figure 4.2 : Hourly Profiles of Goods Vehicle Trips

4.3 Trip Journey Time

4.3.1 The goods vehicle trip journey time distribution is depicted in **Figure 4.3**. It indicates that close to 74% of the trips took 30 minutes or less to travel from origin to destination; 22% took more than 30 minutes to an hour; and about 4% took more than an hour. The mean journey time was 27 minutes.

4.3.2 Further analysis by vehicle type reveals that trips made by van-type LGVs and LGVs were generally shorter with a mean journey time of 24 to 25 minutes. HGV and container vehicle trips were longest. The estimated mean journey times for HGVs and container vehicle trips were 34 and 41 minutes respectively.

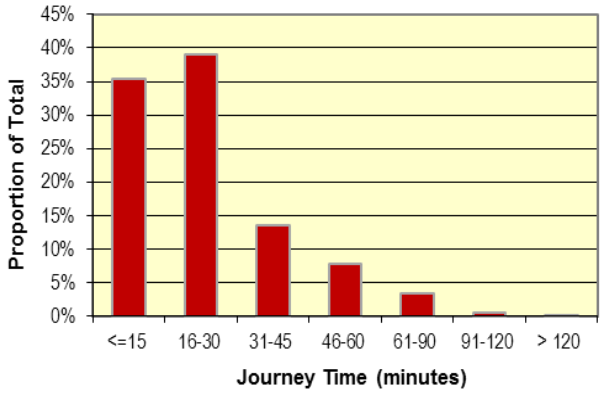


Figure 4.3 : Distribution of Trip Journey Time

4.4 Trip Movements

4.4.1 The estimated daily goods vehicle trip movements on a normal weekday are summarised in **Table 4.3** by main geographical region.

Table 4.3 Major Trip Movements

To/ From	HKI	Kowloon	NT	BCP	Total
HKI	77,500	21,500	10,200	100	**109,300**
Kowloon	19,400	166,900	72,200	500	**259,000**
NT	8,500	71,000	317,200	10,800	**407,500**
BCP	100	800	8,700	N/A	**9,700**
Total	**105,500**	**260,300**	**408,400**	**11,400**	**785,500**

Legend: "HKI" denotes "Hong Kong Island"; "NT" denotes "New Territories"; "BCP" denotes "Boundary Control Point"

4.4.2 As indicated in the table above, the goods vehicle trips were largely dominated by intra-regional movements, among which the intra-NT movement accounted for the largest proportion (40%) of the total number of daily trips, which was followed by intra-Kowloon (21%) and intra-HKI (10%). These three intra-regional movements accounted for over 70% of the daily trip total. A similar phenomenon was observed from the GVTCS 2003.

4.4.3 Cross-boundary trips constituted less than 3% of the daily goods vehicle trip total, and was close to 4% in the GVTCS 2003.

4.4.4 More detailed analysis of the goods vehicle trip origin and destination locations by broad district and by vehicle type are given in **Table A.2** in the Appendix. As shown in the table, the most common origin/destination district for goods vehicle trips was Kwai Chung. About 9% of the total daily goods vehicle trips originated from Kwai Chung, and the same applied to the destination end. This was attributed to the frequent activities of container vehicles and other heavy goods vehicles serving the Container Terminal and other backup facilities located in the district. For van-type LGVs and LGVs, their trip ends were more dispersed

4.5 Loading Conditions

4.5.1 The loading conditions of goods vehicles by vehicle type are summarised in **Figure 4.4**. HGV trips showed the largest proportion (36%) of "full" loading (defined as carrying more than 80% of the loading capacity) followed by container vehicle (25%) and MGV (20%).

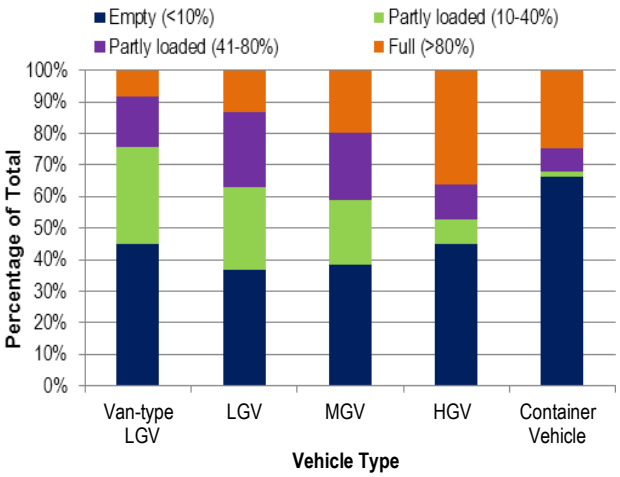


Figure 4.4 : Loading Conditions by Vehicle Type

4.5.2 In general, some 40% or more of trips were made by "empty" goods vehicles (defined as carrying less than 10% of the loading capacity". Container vehicle trips showed the largest proportion (66%) being empty-loaded.



Kwai Tsing Container Terminal is a key special generator of goods vehicle traffic

4.6 Usage of Toll Roads/Tunnels

4.6.1 Most (76%) of the trips reported did not involve the use of any toll road/tunnel. Among the various tunnels, Cross Harbour Tunnel was the most frequently used. 5% of the surveyed trips included the use of Cross Harbour Tunnel, followed by Tseung Kwan O Tunnel (3%), Lion Rock Tunnel (3%) and Shing Mun Tunnel (3%). The remaining (10%) of the surveyed trips used other toll roads/tunnels.

5. OTHER SURVEY RESULTS

5.1 Goods Vehicle Trip Generation Survey

5.1.1 Traffic count data was collected on goods vehicle trips produced from or attracted to specific land uses throughout the day. Based on the count data together with the surveyed site information, trip generation by land use type are derived.

5.1.2 Illustrative trip generation on a territory-wide basis are shown in **Table 5.1**. **Table A.3** in the Appendix provides further breakdowns of the survey results by Hong Kong Island, Kowloon and the New Territories.

Table 5.1 Weekday Daily Goods Vehicle Trip Production and Attraction Rates

Land Use	Unit	Whole Territory	
		Pro	Att
Education			
Kindergarten	per site	4.875	4.875
Primary	per 100m^2 GFA	0.206	0.206
Secondary	per 100m^2 GFA	0.219	0.221
Tertiary	per 100m^2 GFA	0.085	0.085
Domestic			
Private Residential	per flat	0.059	0.058
Public Housing	per flat	0.030	0.031
Subsidised Sale Flats	per flat	0.028	0.028
Non-Domestic			
Retail	per 100m^2 GFA	1.766	1.790
Hotel	per 100m^2 GFA	0.099	0.100
Industrial	per 100m^2 GFA	0.609	0.615
Office	per 100m^2 GFA	0.331	0.321

Legend: Pro – Trip Production; Att – Trip Attraction
GFA – Gross Floor Area

5.2 Goods Vehicle Composition Survey

5.2.1 The data collected has been used to establish detailed screenline/cordon goods vehicle flows to facilitate the GVIS and SGIS trip expansion process. Summary results are provided in **Table A.4** in the Appendix.

6. COMPARISON OF GVTCS 2011 RESULTS WITH PREVIOUS RECORDS

6.1 Comparison of Key Findings

6.1.1 The following aims to provide a summary comparison of how the goods vehicle trip-making patterns and characteristics in 2011 had changed since 2003.

6.1.2 **Table 6.1** summarises these comparisons which cover those key parameters on goods vehicle fleet, operation, overnight parking, vehicle utilisation and overall trip-making characteristics.

Goods Vehicle Fleet

6.1.3 The total goods vehicle fleet size including the total licenced goods vehicles and government goods vehicles had decreased by a slight 1% between 2003 and 2011. In comparison, the proportion of van-type LGVs and LGVs in the goods vehicle fleet had increased slightly to 63% in 2011.

6.1.4 As for vehicle ownership, the proportion of goods vehicles owned by single-fleet owners has substantially increased from 39% in 2003 to 47% in 2011. This suggests a structural change in the goods vehicle operation over the 8-year period with more goods vehicles being operated by smaller scale operations and individuals (particularly for van-type LGVs and LGVs).

Operational Characteristics

6.1.5 Similar to 2003, "Transporting Own Goods" and "Providing Transport Services" remained the two main usages in 2011, accounting for 49% and 36% of the total goods vehicles. These compared to the respective percentages of 47% and 40% in 2003, indicating a slight shift of goods vehicles being used by owners/operators for transporting their own goods, rather than providing transport services for others.

6.1.6 A noticeable decrease of "Hiring Out" from 6% in 2003 to 2% in 2011 was observed.

Overnight Parking

6.1.7 The general geographical distribution of overnight parking was consistent in 2003 and 2011. From 2003 to 2011, parking in Kowloon has increased by 3%, while a corresponding decrease in parking in the NT was recorded.

Vehicle Utilisation

6.1.8 The average working hours per year per vehicle has an increase from 2,420 hours per year in 2003 to 2,494 hours per year in 2011.

Trip-making Characteristics

6.1.9 The total goods vehicle trips made within Hong Kong on a regular weekday has decreased by 6%, from 833,000 in 2003 to 785,000 in 2011. Van-type LGV constituted the largest proportion of goods vehicle trips since 2003, accounting for 36% of the daily goods vehicle trips in 2003, and increasing to 40% in 2011. The daily container vehicle trips showed a noticeable decrease from 85,000 in 2003 to 67,000 in 2011.

6.1.10 The overall goods vehicle trip rate decreased from 7.4 trips per day in 2003 to 7.0 trips per day in 2011. This is in line with the reduction in the total number of goods vehicle trips as the goods vehicle fleet size remained more or less the same between 2003 and 2011. Among different vehicle types, the daily trip rates generally decreased with the exceptions of HGV and container vehicle which showed marginal increases in the daily trip rate per vehicle.

6.1.11 The distribution of trip ends by main region remained relatively constant between 2003 and 2011. In 2011, New Territories (excluding Tsuen Wan/Kwai Tsing and boundary control points) accounted for the largest proportion (34%) of trip ends, closely followed by Kowloon (33%) and Tsuen Wan/Kwai Tsing (18%). These compared to 36%, 35% and 14% respectively in 2003. More notable increase was observed in the percentage of goods vehicle trips to and from Tsuen Wan/Kwai Tsing, possibly due to the continuous growth in the Container Terminal (CT) – related traffic, especially after the opening of CT9.

6.1.12 With the exception of van-type LGV, the Value of Time (VoT) for other vehicle types has increased in real terms from 2003 to 2011. The largest increase was with container vehicle (34%), followed by HGV (21%). The VoT for van-type LGVs went down by 9% in real terms (albeit a 7% increase in nominal terms) over the 8-year period. The overall real increase was about 18% with all goods vehicle types combined in real terms.

6.1.13 The average goods vehicle operating costs (inclusive of all costs including capital cost of vehicle) per kilometre travelled decreased from HK$13.1 (in 2011 prices) in 2003 to HK$9.2 (in 2011 prices) in 2011, a notable 30% reduction from 2003 to 2011 in real terms among all goods vehicles. This is a result of the drop in annual expenses incurred on goods vehicles as discussed earlier in Section 3.4 of this report (also refer to Table 3.4), coupled with the increase in average daily distance travelled as also shown in **Table 6.1**.

Table 6.1 Comparison of Key Goods Vehicle Trip-making Characteristics in 2003 and 2011

Key Parameters	2003[1]	2011
Goods Vehicle Fleet		
Total No. of Goods Vehicles	113,000	111,946
Proportion of Van-type LGVs and LGVs	61%	63%
Proportion of Goods Vehicles Owned by Single Fleet Owners (excl. Government Vehicle)	39%	47%
Operational Characteristics		
Main Usage of Goods Vehicle:		
Transport Own Goods	47%	49%
Providing Transport Services	40%	36%
Hiring Out	6%	2%
Overnight Parking		
Distribution of Overnight Parking Location in Hong Kong:		
Hong Kong Island	8%	8%[2]
Kowloon	18%	21%[2]
New Territories	74%	71%[2]
Vehicle Utilisation		
Average Working Hours per Year (hrs)	2,420	2,494
Annual Average Daily Distance Travelled (km)	87	105[3]
Trip-making Characteristics		
Goods Vehicle Trips by Vehicle Type per Weekday:		
Van-type LGV	304,000 (36%)	316,000 (40%)
LGV	235,000 (28%)	211,000 (27%)
MGV	183,000 (22%)	164,000 (21%)
HGV	26,000 (3%)	28,000 (4%)
Container Vehicle	85,000 (10%)	67,000 (9%)
Total	833,000 (100%)	785,000 (100%)
Overall Goods Vehicle Trip Rate per Weekday (Trips/Vehicle)	7.4	7.0
Distribution of Trips by Time of Day:		
0700-1900	90%	94%
2200-0400	3%	1%
Distribution of Trips Ends by Region:		
Hong Kong Island	13%	14%
Kowloon	35%	33%
Tsuen Wan / Kwai Tsing	14%	18%
Remaining NT	36%	34%
Boundary Control Points	2%	1%
Value of Time ($/min) (in HK$ at 2011 prices) [4]:		
Van-type LGV	2.02	1.83
LGV	2.07	2.11
MGV	2.40	2.75
HGV	2.64	3.21
Container Vehicle	2.26	3.04
Average Vehicle Operating Cost per Kilometre Travelled [5] (2011 prices) [4]	13.1	9.2

Notes:
(1) The 2003 figures presented were extracted from GVTCS 2003, Final Technical Report.
(2) Based on locations where the goods vehicles were most usually parked when not in operation in the 6 months before enumeration as cited by the goods vehicle owners. These could however vary from the locations where the GVs were actually parked overnight on the reference travel day.
(3) Estimated by dividing the annual kilometres travelled by 365 days.
(4) Converted from 2003 to 2011 prices based on Composite Consumer Price Index growth (+18.2%) between 2003 and 2011.
(5) Inclusive of all costs including amortised vehicle capital cost.

7. NEXT STEP

7.1 Application of Data for Model Enhancement

7.1.1 One of the main objectives of the GVTCS 2011 is to collect up-to-date goods vehicle trip characteristics data and information to facilitate the continuous enhancement of the CTS transport model. The current Goods Vehicle Trip Model within the CTS model was developed based on the GVTCS 2003 data.

7.1.2 The mainstay of the Goods Vehicle Trip Model development will be the goods vehicle trip data extensively collected from the GVIS and SGIS. Data collected through other surveys of GVTCS 2011 will also provide valuable supplementary information as reference. All these data will form the basis for subsequent re-calibration and enhancement of the CTS model in the next stage of the Consultancy Study.



Up-to-date trip characteristics data of goods vehicles collected by GVTCS 2011 will be applied for subsequent enhancement of the CTS model